Class 10 – Science 2

Chapter 1: Heredity and Hereditary Changes

Topic: Evolution (Easy, Exam-Oriented Notes

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1. Heredity

Heredity is the transfer of biological characters from one generation to the next generation through **genes**.

Examples of hereditary characters:

- Height
- Skin colour
- Eye colour

Genes carry information from parents to offspring.

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2. Scientists and Their Contributions

Johann Gregor Mendel (1866)

- Known as the **Father of Genetics**
- He studied heredity using pea plants
- His research laid the foundation of modern genetics

Hugo de Vries (1901)

- Proposed the **Mutation Theory**
- Explained that **sudden changes** in organisms occur due to mutations

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Walter and Sutton (1902)

- Observed **paired chromosomes** in grasshopper cells
- Proved that **genes are present on chromosomes**

Avery, MacLeod and McCarty (1944)

- Proved that **DNA is the genetic material** in all living organisms
- Exception: **Viruses**

Jacob and Monod (1961)

- Proposed a model of **protein synthesis**
- Helped in understanding the **genetic code** in DNA
- Their work led to the development of **recombinant DNA technology**

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3. Importance of the Science of Heredity

The science of heredity is useful for:

- Diagnosis, treatment and prevention of **hereditary diseases**
- Production of **hybrid plants and animals**
- Use of microbes in **industrial processes**
- Development of **genetic engineering**

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4. Central Dogma

The flow of genetic information in a cell is called the **Central Dogma**.

DNA → RNA → Protein

- DNA stores information for protein synthesis
- Proteins control body structure and functions

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5. Role of DNA and RNA

- DNA contains information for making proteins
- RNA helps in transferring this information
- Proteins are essential for growth and functioning of the body

6. Transcription

Transcription is the process of synthesis of RNA from DNA.

Important points:

- Only **one strand of DNA** is used
- **mRNA** is formed
- The nucleotide sequence of mRNA is **complementary** to DNA
- RNA contains **uracil (U)** instead of **thymine (T)**

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1. mRNA and Genetic Code

- The **mRNA** formed in the **nucleus** moves into the **cytoplasm**.
- mRNA carries the **coded message** from DNA.
- This message contains codes for **amino acids**.
- Each amino acid is coded by **three nucleotides**.
- This three-nucleotide code is called a **Triplet Codon**.

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2. Triplet Codon & Scientist

- **Dr. Har Gobind Khorana** (scientist of Indian origin):
  - Discovered **triplet codons** for **20 amino acids**
  - Awarded the **Nobel Prize in 1968**
  - Shared with two other scientists

📌 *Very important for 1–2 mark questions*

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3. Translation

Translation is the process of **protein synthesis** using the message present on mRNA.

Important points:

- Each mRNA contains **thousands of triplet codons**
- **tRNA** brings the required amino acids
- tRNA has an **anticodon**
- Anticodon is **complementary** to the codon on mRNA
- Amino acids are joined together by **peptide bonds**
- **rRNA** helps in this bonding

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4. Translocation

- During translation, the **ribosome moves** along the mRNA
- It moves by a distance of **one triplet codon at a time**
- This movement of ribosome is called **Translocation**

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5. Formation of Proteins

- Many amino acid chains combine to form **complex proteins**
- Proteins:
  - Control **body functions**
  - Decide the **appearance** of living organisms

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6. Genes and Heredity

- Living organisms produce offspring **due to genes**
- Some genes are transferred to the next generation **without change**
- Because of this, **parental characters** appear in offspring

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7. Mutation

- Sometimes **sudden changes** occur in genes
- Change in the position of a nucleotide in a gene is called **Mutation**
- Mutation may be:
  - **Minor**
  - **Major**

Example:

- **Sickle cell anemia** is caused due to mutation

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8. Importance of Mutation

- Mutation is a **continuous (everlasting) process**
- It is one of the **evidences of Darwin's theory of natural selection**

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1. Evolution – Definition

Evolution is the **slow and gradual change** that occurs in living organisms over a **long period of time**.

- It is a **very slow process**
- It leads to the **development of new organisms**
- Formation of **new species** due to changes in characters over many generations is called **evolution**
- These changes occur due to **natural selection**

📌 *Exam definition line*
👉 Formation of new species due to gradual changes over generations as a response to natural selection is called evolution.

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2. Origin of Life on Earth

- About **3.5 billion years ago**, life did **not exist** on Earth
- Initially, only **simple elements** were present in the oceans
- From these, **simple organic and inorganic compounds** were formed
- Over a long time, **complex compounds** like:
  - Proteins
  - Nucleic acids

  were formed

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3. Formation of First Living Cells

- Mixture of organic and inorganic compounds led to formation of:
    - **Primitive cells**
- These cells:
    - Increased in number using surrounding chemicals
    - Showed **variations**
- According to **natural selection**:
    - Cells that could **adjust and grow** survived
    - Cells that could not adjust **perished**

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4. Diversity of Living Organisms

- Today, **crores of species** of plants and animals exist on Earth
- There is **huge diversity** in shape and complexity

Animal diversity:

- From **unicellular Amoeba and Paramecium**
- To **human beings and giant whales**

Plant diversity:

- From **unicellular Chlorella**
- To **large banyan tree**

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5. Distribution of Life on Earth

- Life exists:
    - From **equator to poles**
    - In **air, water, land and rocks**
- Humans have always been curious about:
    - **Origin of life**
    - **Reason for diversity** of organisms

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6. Theory of Evolution

- Among many theories, the theory of **gradual development of living organisms** is accepted

According to the Theory of Evolution:

- First living material (**protoplasm**) was formed in the **ocean**
- Then **unicellular organisms** appeared
- Gradually:
    - Unicellular → Multicellular organisms
    - Simple → Complex organisms
- All these changes were:
    - **Slow**
    - **Gradual**
- Total duration of evolution is about **300 crore years**

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7. Nature of Evolution

- Evolution is:
    - **Gradual**
    - **Progressive**
    - **Multi-dimensional**
- Changes occurred in:
    - Structure
    - Function
    - Organization of organisms

📌 **Definition to remember:**
👉 Progressive development of plants and animals from simpler ancestors having different structural and functional organization is called evolution.

Class 10 – Science 2

Chapter 1: Heredity and Hereditary Changes

Topic: Evidences of Evolution (Easy, Exam-Oriented Notes)

Evidences of Evolution

All theories of evolution suggest that evolution is a continuous (everlasting) process.

To support this, various evidences are available.

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1. Morphological Evidences

Meaning:

Morphology means external structure or appearance of organisms.

Explanation:

- Many similar external features are observed in animals and plants.

Examples:

In animals:

- Structure of mouth
- Position of eyes
- Structure of nostrils
- Ear pinnae
- Hair on the body

In plants:

- Shape of leaves
- Leaf venation
- Leaf petiole

Conclusion:

- Such similarities indicate that:
  - Organisms have similar characteristics
  - They have originated from a common ancestor

📌 *Exam line:*

👉 Morphological similarities prove common origin of organisms.

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2. Anatomical Evidences

Meaning:

Anatomy refers to the internal structure of organisms.

Explanation:

- At first glance, the following organs look different externally:
  - Human hand
  - Cat's foreleg
  - Whale's flipper
  - Bat's patagium (wing)
- Their functions are different, but:
  - Bone structure
  - Arrangement of bones are similar in all.

Conclusion:

- Similar internal structure indicates that:
  - These organs are homologous organs
  - Animals have evolved from a common ancestor

📌 *Very important exam point*

Key Points to Remember

- Morphological evidence → similarity in external features
- Anatomical evidence → similarity in internal structure
- Both support the theory of evolution

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3. Vestigial Organs

Definition:

Vestigial organs are degenerated or underdeveloped organs which are useless in a particular organism.

Explanation:

- New organs cannot develop suddenly to suit a changing environment.
- Instead, existing organs change gradually over a long time.
- An organ which is useful in one condition may become useless or harmful in another condition.
- According to natural selection, such organs start degenerating.
- It takes thousands of years for an organ to disappear completely.
- Vestigial organs may be seen in different stages of degeneration in different animals.
- An organ which is vestigial in one organism may be functional in another organism.

Examples:

- Appendix:
  - Useless in humans
  - Useful and functional in ruminants
- Ear pinna muscles:
  - Useless in humans
  - Useful in monkeys
- Vestigial organs in humans:
  - Tail bone (coccyx)
  - Wisdom teeth
  - Body hairs

📌 *Very important for exam*

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4. Paleontological Evidences

Meaning:

Paleontology is the study of fossils.

Fossils:

- Remnants or impressions of organisms buried under the Earth
- Formed due to natural disasters like:
  - Floods
  - Earthquakes
  - Volcanoes
- Fossils provide information about:
  - Organisms that lived millions of years ago

Carbon Dating Method

- After death, carbon intake stops in plants and animals
- Radioactive Carbon-14 (C-14) starts decaying
- Carbon-12 (C-12) is non-radioactive and remains constant
- The ratio of C-14 to C-12 changes over time
- By measuring radioactivity of C-14, age of fossil can be calculated

📌 This method is called Carbon Dating

Uses of Carbon Dating

- Used in:
  - Paleontology
  - Anthropology
- Helps to find:
  - Age of human fossils

- Age of manuscripts

Conclusion from Fossil Study

- Once age of fossils is known:
  - Information about past organisms can be understood
- Fossil records show that:
  - Vertebrates evolved gradually from invertebrates

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5. Connecting Links

Definition:

Organisms which show morphological characters of two different groups are called connecting links.

Examples:

Peripatus

- Shows characters of Annelida:
  - Segmented body
  - Thin cuticle
  - Parapodia-like organs
- Also shows characters of Arthropoda:
  - Tracheal respiration
  - Open circulatory system

📌 Conclusion:

👉 Peripatus is a connecting link between Annelida and Arthropoda

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Duck-billed Platypus

- Lays eggs like reptiles
- Has mammary glands and hair like mammals

📌 Conclusion:
👉 Platypus is a connecting link between reptiles and mammals

Lung Fish

- Is a fish but performs respiration using lungs

📌 Conclusion:
👉 Shows that amphibians evolved from fishes

Overall Conclusion from Connecting Links

- Mammals evolved from reptiles
- Amphibians evolved from fishes

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6. Embryological Evidences

Meaning:

Study of embryonic development is called embryology.

Explanation:

- Comparative study of embryos of different vertebrates shows:
  - Great similarities in early stages
  - Differences appear in later stages
- These similarities gradually decrease as development proceeds

📌 Conclusion:

👉 Similarity in early embryonic stages indicates common origin of vertebrates.

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7. Darwin's Theory of Natural Selection

Scientist:

Charles Darwin

Book:

"Origin of Species"

Main Points of Darwin's Theory

- Organisms reproduce in large numbers
- There is struggle for existence among organisms
- Organisms compete for food, space, and survival
- Only organisms with useful variations survive
- Nature selects the fittest organisms
- Selected organisms reproduce and give rise to new species

📌 This is called "Survival of the fittest"

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8. Objections to Darwin's Theory

- Natural selection is not the only factor of evolution
- No explanation for useful and useless variations
- No clear explanation for:
  - Slow changes
  - Sudden (abrupt) changes

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9. Importance of Darwin's Work

- Despite objections, Darwin's theory:
  - Is a milestone in evolution
  - Laid the foundation for modern evolutionary biology

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1. Lamarckism

Scientist:

Jean-Baptiste Lamarck

Main Idea:

Lamarck proposed that morphological changes in organisms lead to evolution.

- These changes occur due to:
  - Use of organs
  - Disuse of organs
- This principle is called "Use and Disuse of Organs"

Inheritance of Acquired Characters

- Characters acquired during lifetime of an organism are transferred to the next generation
- This concept is called Inheritance of Acquired Characters
- Lamarck's theory is also called Lamarckism

Examples of Lamarckism

- Giraffe:
  - Neck became long due to stretching to eat leaves of tall trees

- Ironsmith:
    - Strong shoulders due to continuous hammering
- Ostrich and Emu:
    - Weak wings due to disuse
- Swan and Duck:
    - Webbed legs useful for swimming
- Snakes:
    - Lost legs due to burrowing habit

📌 According to Lamarck, all these are acquired characters passed to next generation.

Rejection of Lamarckism

- Development or degeneration of organs due to activity was accepted
- But transfer of acquired characters was rejected
- Experiments proved that:
    - Changes acquired during lifetime are not inherited
- Hence, Lamarck's theory was disproved

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2. Speciation

Definition:

Formation of new species from existing species is called speciation.

Species:

- Group of organisms that can:
    - Reproduce naturally
    - Produce fertile offspring

Causes of Speciation

- Genetic variations
- Geographical isolation
- Reproductive isolation
- Differences in:
    - Habitat
    - Food
    - Reproductive period

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3. Human Evolution

- Present biodiversity evolved from simple unicellular organisms
- Human evolution is a part of this long evolutionary process

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Major Stages in Human Evolution

- Dinosaurs became extinct about 7 crore years ago
- Monkey-like animals evolved from lemur-like ancestors
- Loss of tail occurred about 4 crore years ago
- Ape-like animals evolved with:
    - Larger brain
    - Better hands

Evolution of Apes

- Gibbon and Orangutan evolved in Asia
- Gorilla and Chimpanzee evolved in Africa about 2.5 crore years ago

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Evolution of Human-like Animals

- Forests declined → apes started living on land
- Erect posture developed
- Hands became free for use
- First human-like animals evolved about 2 crore years ago

Early Humans

- Ramapithecus:
  - First human-like fossil (East Africa)
- Australopithecus:
  - Evolved about 40 lakh years ago
- Genus Homo:
  - Appeared about 20 lakh years ago
- Homo erectus:
  - Fully erect posture
  - Lived about 15 lakh years ago
  - Found in China and Indonesia

Wise Man – Homo sapiens

- Use of fire about 1 lakh years ago
- Neanderthal man:
  - First wise man
- Cro-Magnon man:
  - Evolved about 50,000 years ago
- Agriculture started about 10,000 years ago
- Writing developed about 5,000 years ago
- Modern science developed about 400 years ago
- Industrial society developed about 200 years ago
